# LUVION: The MPC-Driven Sovereign Asset Fortress

**Technical Whitepaper (Draft)**

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## Abstract

Luvion is the world's first security middleware that integrates **post-quantum algorithms** with a **decentralized revocation protocol**. Through **distributed threshold signing (MPC)** and the **L-SG revocation mechanism**, we build a "last 24 hours" safety net for Web3 assets, eliminating the risk of total loss from single-point private-key failure.

The protocol is an enterprise-grade, non-custodial asset layer that replaces the single private-key paradigm with a **distributed-compute-based sovereignty** model. By combining **Threshold MPC** with **PQC** and **ZK-Fault Proofs**, Luvion eliminates single points of failure, hardens against quantum adversaries, and provides verifiable integrity of every signature shard. A **self-healing node mesh** with proactive resharing and ZK-verified shard claims sets a new standard for institutional-grade digital asset security.


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## 1. Introduction

### 1.1 Motivation

Digital asset custody today is dominated by **single-point-of-failure** architectures: a single private key, or a small set of keys, controls unbounded value. Compromise of that key—through malware, social engineering, or future cryptanalytic advances—results in total loss. Institutions and high-net-worth users require a **security paradigm shift**: from "who holds the key" to "how the key is computed and verified."

### 1.2 Design Goals

- **No single key material**: No node (including the user's host) ever possesses a full signing key.
- **Quantum resistance**: Primary authorization layer resilient to Shor-capable adversaries (e.g. NIST-standardized PQC).
- **Verifiable shard integrity**: Each signature shard is accompanied by a zero-knowledge proof of correct computation (ZK-Fault Proofs).
- **Resilience and self-healing**: Node churn and network partitions trigger automatic resharing and consensus recovery.
- **Institutional safeguards**: Time-locked recovery (L-SG), risk-preview engines, and hardware-backed local hardening when the mesh degrades.

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## 2. Problem Statement

### 2.1 Single Point of Failure

Traditional wallets and many institutional custodians rely on **monolithic key storage**. The private key exists in full at least at one location (HSM, enclave, or hot wallet). Exfiltration of that key implies **total compromise**. Threshold schemes that merely split a key into *n* shares still suffer from: (i) reconstruction at signing time (creating a transient full key), and (ii) lack of cryptographic proof that each share was computed correctly within the protocol.

### 2.2 Quantum Threat

**Shor's algorithm** (and variants) threaten all widely deployed **elliptic-curve and RSA-based** cryptography. Once large-scale quantum computers exist, current key agreement and digital signatures become insecure. **Post-quantum migration** must be designed into the authorization and key-derivation layers from the outset, not retrofitted.

### 2.3 Consensus and Liveness Risk

Centralized or low-node-count custody schemes are highly vulnerable to **DDoS and targeted outages**: once the signing quorum is unreachable, the system halts and user operations freeze. Luvion's **33-node dynamic committee** and **liveness-dependent lock** ensure that consensus-layer attacks cannot shorten the revocation window—they can only prolong the locked state, preserving user recovery time.

### 2.4 Trust and Verifiability

In a distributed signing system, **malicious or faulty nodes** may submit invalid shards. Without cryptographic verification, the coordinator cannot distinguish honest computation from forgery. **Zero-knowledge proofs** allow the network to verify that each shard was produced by a valid circuit execution—without revealing the shard's secret content—enabling **fault isolation** and **malicious-node reporting**.

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## 3. Solution Overview

Luvion implements a **sovereign asset fortress** with five pillars:

1. **Threshold MPC (33/22)**
   The "key" is never materialized. Signing requires a **quorum of 22 from 33** geographically and logically distributed nodes (DVC). No single node (including the user device) holds a full key; reconstruction is performed only in a **secure multi-party protocol** that outputs a signature, not a key.

2. **Post-Quantum Hardening (PQC)**
   The primary authorization and key-establishment layer uses **NIST-standardized post-quantum primitives** (e.g. **Kyber-768** for KEM). This ensures that even if an adversary gains classical access to the system, future quantum capability cannot retroactively break the authorization design.

3. **ZK-Fault Proofs**
   Every signature shard submitted to the coordinator is accompanied by a **Groth16 ZK-SNARK proof** over a fixed circuit (e.g. on **Bn254**). The verifier checks the proof and public inputs; failure triggers **malicious-node reporting** and exclusion from the signing set. Thus, **shard integrity** is enforced without exposing secret material.

4. **Self-Healing Mesh**
   A **heartbeat monitor** periodically scans the 33-node mesh. When a node is detected offline or unresponsive, the protocol triggers **proactive resharing** so that the effective threshold remains satisfiable with the live set. Combined with P2P discovery (e.g. **Kademlia**) and overlay broadcast (e.g. **Gossipsub**), the mesh maintains liveness and consistency.

5. **Institutional Safeguards**
   **L-SG (SafeGuard™)** provides time-locked, physical-seed recovery (e.g. QR-code shards). A **risk-preview engine** performs real-time transaction simulation to block high-risk contract interactions.

When the number of active peers falls below the threshold, **hardened local storage** (e.g. TPM or Secure Enclave) is used to re-encrypt local shard material.

### 3.2 Dynamic Verification Committee (DVC)

To balance decentralization and execution efficiency, Luvion replaces a fixed committee with a **Dynamic Verification Committee (DVC)**:

- **Staking pool size**: the protocol allows $N \ge 100$ independent nodes to enter a candidate pool by staking $LVN$.
- **Committee sampling**: at each Epoch (≈ 1 hour), the system uses a **Verifiable Random Function (VRF)** to sample **33 active nodes** from the candidate pool to form the on-duty committee.
- **Threshold consensus**: the committee uses a $t=22,\ n=33$ threshold signing scheme. An attacker would need to compromise 22 globally distributed, dynamically changing nodes within a single epoch—statistically infeasible.
- **Performance target**: 33 nodes is a practical "sweet spot" for MPC communication overhead, keeping signing latency stable at $T < 200ms$ under global network conditions.

### 3.3 Trustless Distributed Key Generation (DKG)

Luvion rejects any centralized private-key distribution. The protocol adopts an improved **Pedersen Verifiable Secret Sharing (VSS)**-style DKG:

- **Decentralized generation**: at Genesis or during committee rotation, each node generates a local random polynomial and interacts with peers via a secret-sharing protocol.
- **Secret shares**: the full private key $SK$ never exists in one place; it only exists as mathematical fragments held in each node's protected memory.
- **Public commitments**: via public commitments (and Lagrange interpolation over commitments), participants can verify the validity of peers' contributions without revealing private shares.

### 3.4 View Change & Liveness Recovery

If the current 33-node committee experiences excessive delay or partial outage, the system triggers **View Change** to preserve liveness:

- **Fault tolerance trigger**: if more than $f=11$ active nodes become high-latency or unresponsive, or if the current committee **fails to reach consensus within 5 seconds**, the protocol initiates View Change.
- **Rescheduling**: the candidate pool immediately intervenes to form a new 33-node committee, achieving **second-scale consensus recovery**.

### 3.5 Asynchronous Threshold Waiting

To handle global network latency without blocking on the slowest node, Luvion uses **asynchronous threshold aggregation**:

- **Threshold-first completion**: The signature aggregator **completes as soon as the first 22** valid partial signatures (shares) are collected—it does **not** wait for all 33 responses.
- **Performance**: This masks long-tail latency; **average signing latency** is kept stable within **250ms** under normal conditions.

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## 4. Technical Architecture

### 4.1 Threshold MPC and Sharding

- **Model**: **33 nodes** (DVC), **threshold 22** (33/22). A valid signature is produced only when at least 22 nodes contribute correct partial signatures (or shards) to a **threshold signature scheme**.
- **Key material**: The "master" secret is split using **Shamir-style / DKG** into 33 shards. Each shard is stored at a distinct node; **no node** has access to more than one shard in the intended deployment.
- **Signing path**: The protocol **never reconstructs** the full private key. Signing is performed via **multi-party computation**: each node computes a partial signature over the message; a **combiner** (or distributed protocol) produces the final signature from at least 22 valid partials (see §3.5 asynchronous threshold waiting).
- **BIP44 / HD**: Shard-derived material supports **path-based derivation** (e.g. BIP44) so that different chains (e.g. Ethereum, Solana) can be addressed from the same MPC root without re-sharing.

### 4.2 Post-Quantum Cryptography (PQC)

- **Role**: The **primary authorization layer** (e.g. user authentication, session establishment, or key encapsulation for sensitive metadata) uses **post-quantum primitives**.
- **Choice**: **Kyber-768** (NIST PQC standardization) is designated for **key encapsulation (KEM)**. This provides **IND-CCA2** security and is believed secure against both classical and known quantum attacks.
- **Integration**: PQC keys may protect the distribution of shard material, attestation channels, or recovery seeds. The design ensures that **quantum-safe** and **classical** components are clearly separated so that migration and auditing are straightforward.

### 4.3 ZK-Fault Proofs (Groth16)

- **Objective**: Ensure that every **shard claim** (i.e. a partial signature or commitment submitted by a node) is **cryptographically verified** to have been produced by the correct circuit and inputs, **without revealing** the shard's secret value.
- **Construction**: Each node, when submitting a shard, also submits a **Groth16 ZK-SNARK proof** (over the **Bn254** curve) that attests: "I know a witness (e.g. my shard and internal state) such that the prescribed circuit evaluates to the claimed public output."
- **Verification**: The coordinator (or any verifier) holds a **Prepared Verifying Key (PVK)**. For each shard claim, it runs **Groth16::verify_proof(pvk, proof, public_inputs)**. If verification fails, the node is **reported as malicious** ( `report_malicious_node` ) and excluded; the protocol may trigger **resharing** or replacement.
- **Middleware**: The ZK check is implemented as **middleware** in the **prepare_signing_shards** pipeline: before any signing round, the set of shard claims is verified; only then does the mesh proceed to **warm_up_connection** and signing. This enforces **fail-fast** semantics and prevents faulty or adversarial shards from entering the threshold combination.

### 4.4 Self-Healing Mesh and Proactive Resharing

- **Topology**: 33 nodes (DVC) form a **P2P mesh** with discovery via **Kademlia** (DHT) and broadcast via **Gossipsub** (or equivalent). Shard updates and resharing messages are propagated through this overlay.
- **Heartbeat**: A **heartbeat monitor** runs at a fixed interval (e.g. 5 seconds). It **scans** the mesh (e.g. `scan_node_mesh` ) and marks nodes that do not respond as **offline**.
- **Proactive resharing**: For each node detected **offline**, the protocol invokes **trigger_shard_resharing(node_id)**. This initiates **proactive resharing** so that the remaining live nodes obtain new shards and the threshold (22) remains achievable. No full key is ever reconstructed; resharing is done in a distributed manner.

- **Consensus and liveness**: The system periodically **checks shard consensus** and **redistributes shards** after resharing. This keeps the mesh in a **consistent and available** state despite churn.

### 4.5 Institutional Safeguards

- **L-SG (SafeGuard™) Recovery**: A **time-locked recovery** protocol (default **24h** revocation window) with **physical seed shards** (e.g. QR-code). Recovery requires the time lock to expire and physical shards to be combined in a controlled environment. **Revocation commands** use **Kyber-768** so they remain secure in the quantum era. This mitigates remote coercion and single-device loss.
- **Liveness-Dependent Lock**: When the network status is **not Optimal** (e.g. active nodes < 22, or View Change in progress), the **asset release countdown is suspended**. Attackers cannot shorten the revocation window—they can only prolong the locked state, ensuring users always have sufficient time to revoke.
- **Risk-Preview Engine**: Before any transaction is broadcast, a **sandboxed simulation** (e.g. `cmd_simulate_transaction` ) runs. High-risk patterns are flagged; the user must explicitly confirm or abort. This reduces **approval and phishing** risks.
- **Hardened Local Storage**: When the number of **active peers** drops below the **threshold** (22), the client triggers **hardened_local_storage**: local shard material is **re-encrypted** using **TPM** or **Secure Enclave**. This limits exposure if the mesh is temporarily degraded.
- **Biometric and 2FA**: Critical actions (e.g. withdrawal) require **biometric authentication** (e.g. Touch ID / Face ID) and optional **2FA**, ensuring that even with device access, signing requires user presence.

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## 5. Security Model and Threat Analysis

| Threat | Mitigation |
|---|---|
| **Single node compromise** | No full key; threshold 22/33; ZK verification rejects invalid shards. |
| **Quantum adversary** | PQC (Kyber-768) in authorization; design allows replacement of classical components. |
| **Malicious or faulty shard** | Groth16 ZK-Fault Proof per shard; failure → `report_malicious_node` and exclusion. |
| **Network partition / node churn** | Heartbeat + proactive resharing; mesh self-heals; **liveness-dependent lock** and hardened storage when below threshold. |
| **Phishing / approval abuse** | Risk-preview engine; sandboxed simulation; explicit user confirmation. |
| **Device loss** | L-SG time-lock + physical shards; no single device holds recoverable full key. |

### 5.2 Dynamic Staking Coefficient

To deter low-cost attacks, Luvion applies a dynamic minimum staking requirement for active nodes that scales with secured value (TVL):

[ Stake_{min} = \alpha \cdot \frac{TVL}{N} ]

where (\alpha) is a security coefficient. The objective is to keep the economic cost of misbehavior consistently **above 300%** of the potential profit.

### 5.3 Tokenomics

- **$LVN Staking**: Nodes must stake $LVN to obtain committee selection weight; misbehavior leads to **Slashing**.
- **Dynamic staking coefficient**: Minimum stake scales with secured TVL (§5.2), keeping attack cost above 300% of potential profit.

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## 6. Robustness & Security Hardening (Chapter VI)

### 6.1 Recursive Proof Aggregation

To reduce communication overhead when verifying ZK proofs across large node clusters (33 nodes+), Luvion adopts a **Recursive SNARKs** architecture. Traditional **O(n)** linear verification is replaced with **O(1)** constant-time verification, keeping verification latency under **T < 100ms** even under high concurrency.

### 6.2 Hybrid PQC-ZK State Compression

Post-quantum signatures (Kyber-768 / Dilithium) incur high on-chain Gas cost. Luvion uses a **layered verification** design: full-strength PQC verification remains inside the Mesh; only the verification result is **compressed into a short ZK-SNARK proof** and submitted to L1, **reducing user Gas cost by ~98%**.

### 6.3 Consensus-Gated Slashing

No slashing takes effect on a single authority's say. Any slashing decision must be backed by **BFT consensus** with **Q > (2/3)N + 1** signatures. A randomly selected **Guardian Nodes** governance committee acts as final arbiter, ensuring **social-consensus-level** self-healing.

### 6.4 Quorum & Split-Brain Protection

When the number of **active nodes** falls below the **signature threshold** (*N < 22*), the system enters **protective lock mode** and **rejects all signing requests**. This prevents double-spend under network partition and keeps **global cross-region consistency**.

### 6.5 Multi-channel Async Revocation

To counter DDoS or social-engineering attacks, the **L-SG protocol** supports **smart time-delay**: on anomaly detection, the revocation window is extended to **72 hours**, and pre-configured **trusted circles** can asynchronously trigger emergency lock, providing a **physical last line of defense** for user assets.

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## 7. Technical Roadmap

| Phase | Milestone | Objective |
|---|---|---|
| **I** | **Core MPC** | 33-node DVC threshold; Shamir/DKG; no full-key materialization. |
| **II** | **PQC Hardening** | Kyber-768 integration in authorization and KEM; NIST alignment. |
| **III** | **ZK-Fault Proofs & Audit** | Groth16 (Bn254) verification in `prepare_signing_shards`; third-party code audit; ZKP circuit audit. |

| | | |
|---|---|---|
| **IV** | **Self-Healing & Production** | Heartbeat monitor; proactive resharing; Kademlia/Gossipsub; liveness-dependent lock; hardened storage and L-SG. |

### 7.2 Protocol Evolution Roadmap

- **Phase 1 (Genesis)**: launch 18 core founding nodes to establish the security baseline.
- **Phase 2 (Decentralization)**: enable $LVN$ staking, expand to 100+ nodes, and activate VRF-based DVC (33-node committee).
- **Phase 3 (Scaling)**: introduce sharded signing to support 1800+ nodes in parallel, serving global institutional-grade custody.

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## 8. Conclusion

Luvion is not only a security tool but a necessary step for Web3 toward **institutional- and quantum-grade** evolution. By deeply integrating **33-node distributed consensus** with the **liveness-dependent lock**, the protocol delivers a truly "worry-free" ownership experience.

Luvion proposes a **sovereign asset fortress** where **no single key** exists, **every shard is ZK-verified**, and the **mesh self-heals** under failure. With **threshold MPC (33/22)**, **PQC**, **ZK-Fault Proofs**, and institutional safeguards (L-SG, **liveness-dependent lock**, risk preview, hardened storage), the protocol targets **institutional-grade** security and auditability. This whitepaper provides a technical and structural foundation for further specification, implementation, and third-party review.

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